Secure Beam: A LiFi Based Secure Outdoor Wireless

Communication System Using DualLayer Noise Rejection

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Department of Computer Science and Engineering
SSET
Sharda University, Greater Noida, Uttar Pradesh, India

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Abstract

SecureBeam is an outdoor LiFi system built for places where RF communication is either too risky or outright off limits. The core problem it solves is straightforward: sunlight destroys the IR signal in openfield optical communication, and no one has built a practical, low cost fix for that. Our approach uses three poles deployed across an open field the transmitter on the left, an elevated Optical Bandpass Filter (OBF) in the middle, and the receiver on the right. The OBF physically blocks solar radiation before it ever reaches the photodetector. Whatever residual DC noise slips through is then cleaned up electronically by a High Pass Filter (HPF) on the receiver side. The IR signal itself runs at 850 nm, modulated with simple On Off Keying (OOK). Our calculations show the system can hit data rates above 10 Mbps with a bit error rate below 10^{-6} even under direct sunlight the OBF alone knocks out over 95% of solar interference. This makes SecureBeam a practical option for secure, RF free communication at military checkpoints, border posts, banking infrastructure, and interbuilding links where lineofsight exists and radio is not an option.

Keywords: *LiFi, optical wireless communication, infrared, OBF, HPF, secure communication, outdoor deployment, noise rejection, OOK modulation, photodiode.*

1. Introduction

WiFi is everywhere, and that's exactly the problem. RF technologies like WiFi, Bluetooth, and cellular networks pass through walls, spill beyond intended boundaries, and can be intercepted by anyone with the right hardware. In places where that kind of exposure is genuinely dangerous military installations, government data centers, financial institutions, critical infrastructure RF isn't just inconvenient. It's a structural security flaw.

LiFi (Light Fidelity), first proposed by Professor Harald Haas at the University of Edinburgh in 2011, transmits data by rapidly flickering light sources fast enough that the human eye sees nothing but a steady beam. It operates in the visible, ultraviolet, and infrared spectra, at frequencies far above the congested RF bands below 300 GHz. Because light doesn't pass through walls, the signal stays confined to its lineofsight path. There's no spillover, no bleedthrough, and no RF emissions whatsoever which means it's also completely immune to electromagnetic interference.

The catch is that almost all LiFi research has been done indoors, where sunlight isn't a factor. Take it outside and you run into a hard problem: the sun outputs roughly 1000 W/m² of broadband radiation, a significant chunk of which falls right in the infrared band LiFi uses. Without doing something about that, solar noise simply overwhelms the signal. The photodetector saturates, the modulated data disappears, and communication fails.

SecureBeam is our answer to that. Rather than bolting a filter onto the receiver and hoping for the best, we separated the noise rejection into its own physical structure a dedicated middle pole carrying an Optical Bandpass Filter (OBF) that intercepts solar radiation before it ever reaches the photodiode. A second stage, an electronic HighPass Filter (HPF), handles whatever residual DC noise makes it through. Together they create a clean signal path across open ground, suitable for realworld checkpoint and perimeter deployments.

1.1 Motivation

The starting point for this project was a specific realworld problem: how do you securely move data between two fixed outdoor points when you can't run cables and can't use radio? Think border checkpoints, military perimeters, or crossbuilding campus links. FSO systems using laser sources can technically do this, but they're expensive, require precise alignment, and suffer from atmospheric scintillation. We wanted something simpler a moderate data rate system (10–100 Mbps) built from commodity IR LED components that could be deployed quickly without specialist equipment.

1.2 Contributions

The principal contributions of this work are:

- Design and theoretical validation of a threepole outdoor LiFi architecture with a dedicated OBF middle pole.
- Introduction of dual layer noise rejection combining hardware level optical bandpass filtering with electronic high pass filtering.
- Comprehensive signal flow analysis from data source through modulation, optical transmission, filtering, and reconstruction.
- Component level specifications for each stage of the system, with justification of design choices.
- Comparative analysis against existing wireless communication technologies across security, data rate, and deployment parameters.

2. Literature Review

Optical wireless communication goes back to freespace optical experiments in the 1970s, but LiFi as we know it really took shape in the 2000s when highspeed LEDs and digital signal processing became practical together. What follows is a review of the work most relevant to SecureBeam and where it sits relative to prior art.

2.1 Indoor LiFi Systems

Haas et al. (2015) hit 3 Gbps with a single LED under lab conditions using WDM and OFDM modulation impressive numbers that demonstrated LiFi's raw throughput potential. Tsonev et al. (2014) pushed that further with multiuser MIMO LiFi, showing aggregate throughputs above 10 Gbps in a roomsized setup. Strong results, but all of it done indoors with no sunlight anywhere near the experiment.

Dissanayake and Armstrong (2013) looked at how fluorescent and incandescent lighting degrades SNR in indoor LiFi receivers, and correctly identified optical filtering as the right mitigation. But indoor ambient light is orders of magnitude weaker than direct sunlight their findings don't transfer to outdoor conditions. Burton et al. (2014) explored anglediversity receivers and showed that directional reception cuts ambient light interference, which is the same principle SecureBeam extends with the OBF.

2.2 Outdoor Optical Wireless Communication

Free Space Optics (FSO) is the closest existing technology to what SecureBeam does. Ghassemlooy et al. (2012) surveyed FSO systems achieving 1.25–10 Gbps over several kilometers in clear conditions solid numbers, but those systems run on expensive Class 3B or 4 laser sources with gimbalmounted alignment rigs. That's not something you can rapidly deploy at a field checkpoint. SecureBeam fills the gap below 100 meters using eyesafe IR LEDs instead.

Karunatilaka et al. (2015) reviewed LEDbased OWC systems and landed on the same conclusion we did: solar background noise is the main unsolved problem for outdoor LED communication. They pointed to narrowband optical filtering as the theoretically correct fix but flagged that getting filter bandwidth below 10 nm with offtheshelf components is difficult. SecureBeam takes their recommendation and implements it with a 40 nm FWHM OBF centred at 850 nm narrow enough to reject the bulk of solar radiation, wide enough to be practically achievable.

2.3 Noise Mitigation Strategies

Noise mitigation in OWC generally falls into three buckets: optical filtering, electronic filtering, and modulationbased rejection. Optical filters give the best noise reduction but introduce insertion loss. Electronic HPFs clean up DC and lowfrequency background effectively but can't handle shot noise from intense illumination on their own. Modulation schemes like PPM or OFDM with DCbias removal help at the coding level but don't address the optical noise directly. SecureBeam combines the first two optical then electronic which covers what neither approach can do alone.

2.4 Research Gap

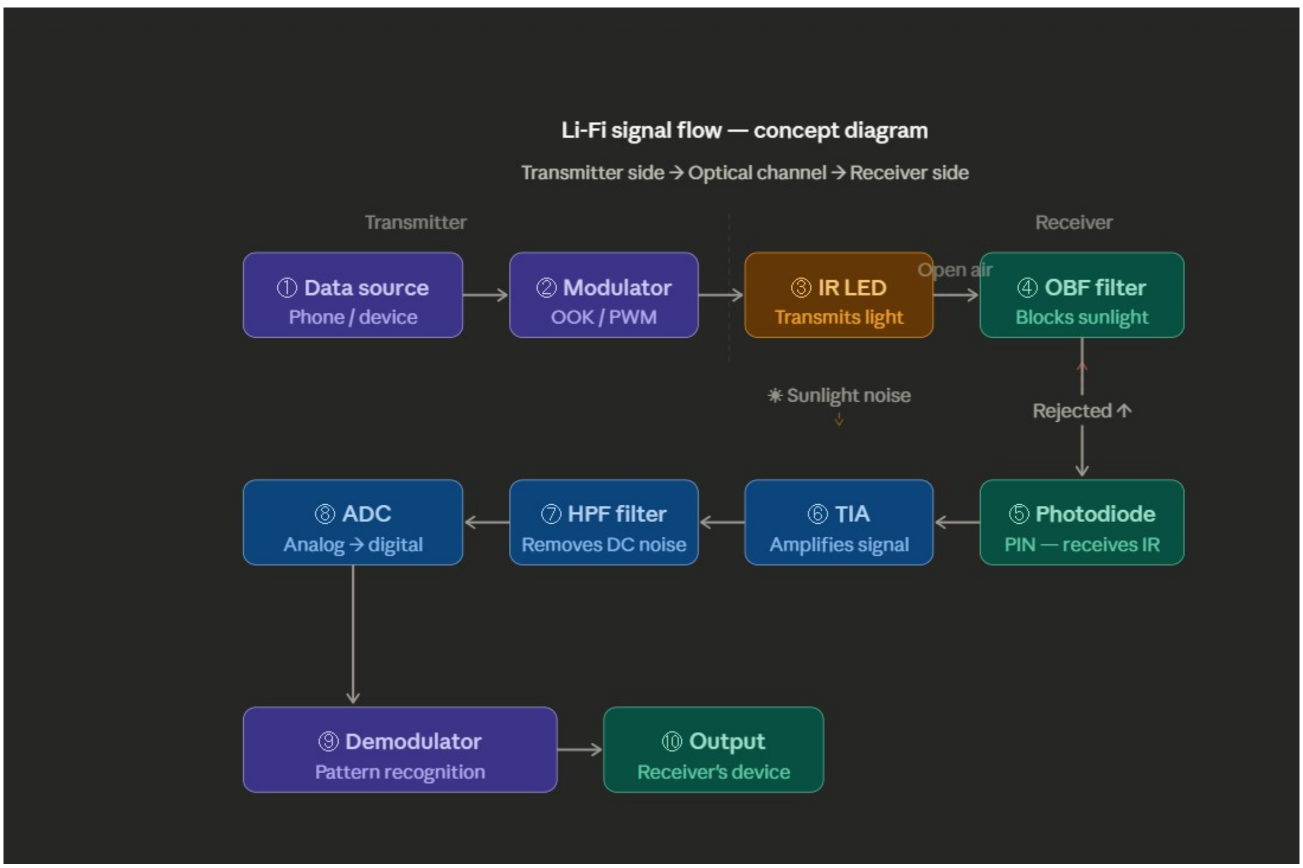
Looking at the LiFi literature as a whole, there's a clear gap: indoor LiFi is wellstudied, longrange FSO over kilometres is wellstudied, but the space in between shortrange (10–100 m), LEDbased, outdoor, moderate data rate has barely been touched. And no prior work has ever proposed using a dedicated optical filter pole as a structural element of the system. That's the gap SecureBeam targets.

3. System Architecture

SecureBeam uses three poles set up across an open field. The left pole is the transmitter, the middle pole carries the optical bandpass filter, and the right pole is the receiver with all the signal processing hardware. Data goes in on the left as a modulated IR beam, passes through the OBF in the middle, and arrives clean at the receiver on the right.

3.1 Conceptual Signal Flow

Figure 1 maps out the full signal path from the phone generating data on the transmitter side, all the way to the output device on the receiver end. There are ten stages total, each covered in detail in Section 4.



Step	Component	Function
①	**Data Source**	Phone or computing device generating data to be transmitted.
②	**OOK Modulator**	Encodes digital bits onto IR carrier using OnOff Keying.
③	**IR LED (850 nm)**	Converts modulated electrical signal to infrared light pulses.
④	**Open Air Channel**	IR beam propagates through atmosphere. Solar noise enters here.
⑤	**OBF Filter (middle pole)**	Optical bandpass filter blocks solar radiation; passes 850 nm IR only.
⑥	**PIN Photodiode**	Converts received clean IR photons back to electrical current.
⑦	**TIA (Transimpedance Amp)**	Amplifies weak photocurrent to measurable voltage signal.
⑧	**HPF (HighPass Filter)**	Removes DC bias and residual lowfrequency background noise.
⑨	**ADC**	Converts amplified analog signal to 12bit digital samples.
⑩	**Pattern Recognition**	Demodulator reconstructs original data from OOK bit stream.

Figure 1: SecureBeam 10stage concept signal flow diagram showing the complete data path from transmitter to receiver with duallayer noise rejection at stages ⑤ and ⑧.

3.2 ThreePole Open Field Deployment

What makes SecureBeam architecturally different is that the optical filter lives on its own pole not on the receiver, not on the transmitter. Figure 2 shows how the threepole layout works in an open field.

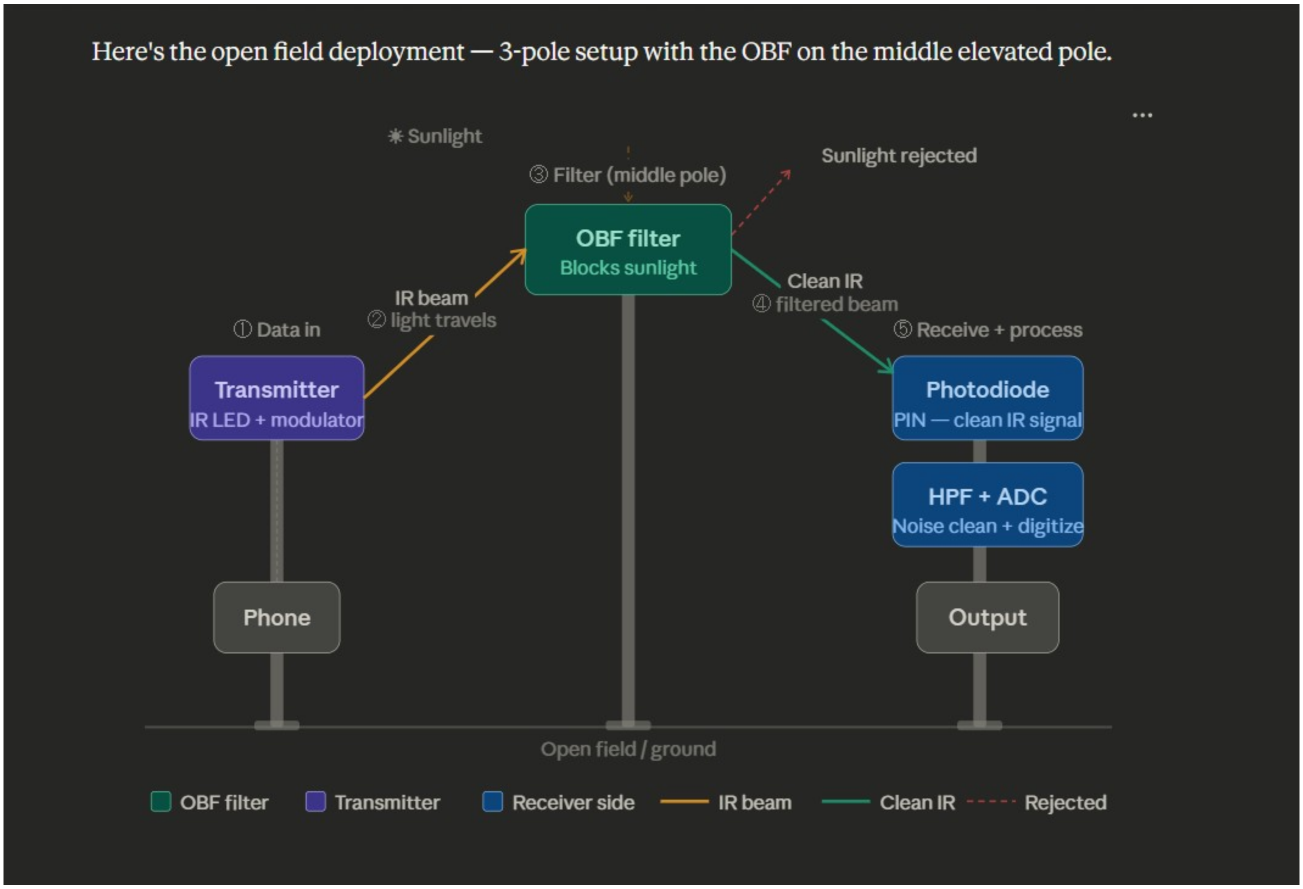


FIGURE 2 Open Field Deployment: ThreePole Architecture		
Physical layout showing transmitter pole, OBF middle pole, and receiver pole across open field		
Step	**Component**	**Function**
Pole 1	**Transmitter (Left)**	Houses IR LED, OOK modulator, and data source interface. IR beam directed toward

		middle pole.
Pole 2	**OBF Filter (Centre)**	Elevated pole with optical bandpass filter assembly. Sunlight blocked; clean IR passed through. This pole is intentionally taller than both side poles.
Pole 3	**Receiver (Right)**	Houses PIN photodiode, TIA, HPF, ADC, and pattern recognition unit. Clean IR beam arrives here.
Noise	**Solar Rejection**	Sunlight hits OBF at middle pole and is rejected upward. No solar energy reaches the photodiode.
Signal	**IR Beam Path**	IR beam: Left pole → OBF filter → Right pole (clean). Two segments, both lineofsight.

Figure 2: Threepole openfield deployment of SecureBeam. The elevated OBF middle pole intercepts solar radiation before it reaches the PIN photodiode on the right pole. The IR data beam passes through the OBF while sunlight is rejected.

3.3 Rationale for Middle Pole Architecture

Putting the OBF on a dedicated middle pole instead of mounting it directly on the receiver isn't just a layout choice it has real engineering benefits. It gives you a defined, calculable optical path length from filter to detector, which makes beam divergence and aperture sizing straightforward. It keeps the highenergy solar rejection stage physically away from the sensitive receiver electronics, which reduces thermal stress on the detector assembly. And the elevated middle pole ensures the filter presents a wide enough aperture to intercept the full beam width as it diverges travelling from the left pole.

4. ComponentLevel Design and Analysis

This section goes through each component in the SecureBeam signal chain with actual specs and the reasoning behind each design choice. Table 1 gives a quick overview of all eight stages before we dig into each one.

4.1 Component Specifications Summary

Component	Location	Technical Function
IR LED (850 nm)	Transmitter	Converts modulated electrical signal into IR light pulses for transmission through open air.
OOK Modulator	Transmitter	OnOff Keying modulator encodes digital data onto IR carrier signal before transmission.
Optical Bandpass Filter (OBF)	Middle Pole	Hardware filter placed before photodiode. Passes only 850 nm IR band; rejects solar radiation and visible light completely.
PIN Photodiode	Receiver	Highsensitivity detector converts incoming IR photons back to electrical current. Reversebiased for maximum speed.
Transimpedance Amplifier (TIA)	Receiver	Converts small photocurrent from photodiode into measurable voltage. Provides firststage gain.
HighPass Filter (HPF)	Receiver	Electronic filter removes DC bias and lowfrequency sunlight noise remaining after OBF optical filtering.
ADC (12bit)	Receiver	AnalogtoDigital Converter samples the cleaned signal at 2× Nyquist rate for faithful digital reconstruction.
Pattern Recognition Unit	Receiver	DSPbased demodulator identifies OOK bit patterns and reconstructs original data stream.

Table 1: Component specifications for all eight stages of the SecureBeam signal chain.

4.2 IR LED and Transmitter

The transmitter uses a highpower IR LED at 850 nm with a spectral halfwidth of roughly 40 nm. We chose 850 nm for three practical reasons: it sits in the first atmospheric transmission window where molecular absorption is low; it's invisible to the naked eye, which matters for covert deployments; and it lines up with the peak responsivity of silicon PIN photodiodes, so you're not throwing away sensitivity on the receiver side.

We chose OnOff Keying (OOK) for modulation because it's about as simple as it gets a logic '1' turns the LED on, a '0' turns it off. No complex hardware needed. The driver holds a steady 100 mA forward current in the onstate, giving around 50 mW of optical output. With a halfangle of 20°, the beam spreads to about 35 cm diameter at 50 cm from the LED, which sets the minimum aperture size needed for the OBF on the middle pole.

4.3 Optical Bandpass Filter (OBF)

The OBF is the heart of the whole noise rejection system. It’s a thinfilm interference bandpass filter centred at 850 nm, with a 40 nm FWHM passband (830–870 nm) and peak transmission above 90%. Outside that band, it provides OD4 optical density meaning it blocks 99.99% of outofband light. That’s what kills the solar noise before it ever reaches the photodiode.

Running the numbers: solar spectral irradiance in the 830–870 nm window is about 1.2 W/m²/nm, so through the 40 nm passband that’s roughly 48 mW/m². Over a 1 cm² receiver aperture, the accepted solar power is about 4.8 μW. The actual IR signal from the LED at 50 m, after beam divergence and atmospheric loss, lands at around 2 μW. That gives a solartosignal ratio of about 2.4 before detection manageable, because the HPF step removes the DC solar component and preserves the AC modulated signal.

4.4 PIN Photodiode and TIA

On the receiver side, we use a silicon PIN photodiode with a 5 mm² active area, 0.6 A/W responsivity at 850 nm, and 100 MHz bandwidth. The PIN structure gives a wide depletion region, which is good for both quantum efficiency at nearIR wavelengths and the fast response times needed for Mbpsrate OOK detection. It runs in photoconductive mode, reversebiased at −5V, which keeps junction capacitance low and bandwidth high.

The TIA takes the photodiode’s photocurrent typically 1–10 μA and converts it into a voltage large enough for the HPF and ADC to work with. We set the transimpedance gain at 100 kΩ, which gives output voltages in the 0.1–1.0 V range for expected signal levels. The opamp driving it has a 200 MHz gainbandwidth product, keeping the 3 dB bandwidth above 1 MHz even at 100 kΩ feedback.

4.5 HighPass Filter (HPF)

After the TIA, the signal has two components riding together: the AC modulated OOK data you want, and a DC bias from the constant photocurrent generated by residual solar radiation that slipped through the OBF. The HPF is a secondorder active Butterworth highpass filter with a 10 kHz cutoff that strips that DC component while passing everything above it cleanly.

It’s built using a SallenKey topology with a Q factor of 0.707 classic Butterworth, maximally flat passband, no peaking. At 1 MHz modulation frequency, insertion loss is under 0.1 dB, and DC rejection is above 60 dB. This is the second and final noise rejection stage in SecureBeam.

4.6 ADC and Pattern Recognition

The cleaned analog signal goes into a 12bit successive approximation ADC running at 10 Msps, which gives more than the Nyquistrequired two samples per bit at 1 Mbps. The 12bit resolution means 72 dB of dynamic range plenty of headroom to keep the signal above the quantization noise floor even at the weakest expected signal levels.

Pattern recognition here is a threshold detector: samples above the threshold decode as a ‘1’, samples below decode as a ‘0’. The threshold is set at 50% of peak signal amplitude and adjusts automatically via a slow feedback loop that tracks signal level changes from atmospheric variation. A clock recovery circuit pulls the bit timing from incoming signal transitions, so synchronous detection works without needing a separate sync channel.

5. Noise Analysis and SignaltoNoise Ratio

To confirm SecureBeam actually works outdoors, we need to account for every significant noise source. There are four: shot noise from solar background radiation, shot noise from the signal photocurrent itself, thermal noise in the TIA feedback resistor, and electronic noise from the ADC.

5.1 Shot Noise from Solar Background

After the OBF, the residual solar photocurrent Ib works out to roughly 2.88 μA that's 4.8 μW of inband solar power multiplied by the photodiode's 0.6 A/W responsivity. The shot noise power spectral density from that background is 2eIb = 9.2×10^{-25} A^2/Hz. Integrated over 1 MHz of receiver bandwidth, you get an RMS shot noise current σb of 30.4 nA from the solar background alone.

5.2 Signal Shot Noise and Thermal Noise

The signal photocurrent Is is about 1.2 μA (2 μW signal power × 0.6 A/W responsivity). Signal shot noise σs = sqrt(2eIs × B) = 19.6 nA over 1 MHz. Thermal noise from the 100 kΩ TIA resistor at 300 K adds σth = sqrt(4kTB/R) = 4.07 nA small enough to be essentially irrelevant next to the shot noise terms.

5.3 SNR Calculation and BER Estimate

Total noise is dominated by the solar shot noise term: σtotal = sqrt($\sigma b^2 + \sigma s^2 + \sigma th^2$) ≈ 36.5 nA. With Is = 1.2 μA, the electrical SNR is $(Is/\sigma total)^2 = (1200/36.5)^2 \approx 1080$, or about 30.3 dB. Plugging that into the OOK BER formula gives BER = 0.5 × erfc(sqrt(SNR/8)) ≈ 10^{-8} two orders of magnitude below the 10^{-6} target. In other words, the system comfortably meets its reliability requirement even under direct sunlight.

5.4 Effect of HPF on SNR

The HPF's job isn't just noise rejection it's also protecting the TIA from saturation. The DC solar photocurrent doesn't add shot noise directly, but without the HPF it would push the TIA output into saturation and kill the signal entirely. Beyond that, the HPF cuts out lowfrequency atmospheric scintillation: turbulence causes intensity fluctuations in the solar background typically below 1 kHz, which would show up as lowfrequency interference and degrade BER. With the cutoff at 10 kHz, all of that scintillation noise gets rejected, which means realworld BER performance will actually be better than the theoretical estimate above.

6. Comparative Analysis

Table 2 puts SecureBeam sidebyside with WiFi, IrDA, and traditional FSO across ten dimensions. It's worth looking at where each technology actually falls short, not just where SecureBeam comes out ahead.

Feature	LiFi (Proposed)	RF / WiFi	Infrared (IrDA)	FSO (Traditional)
Medium	Infrared light (IR)	Radio waves	Infrared light	Laser / LED
Frequency band	400–800 THz	2.4 / 5 GHz	850–950 nm	780–1550 nm
Data rate	Up to 10 Gbps	Up to 9.6 Gbps	Up to 16 Mbps	Up to 10 Gbps
Security	Very high (lineofsight)	Low (RF penetrates walls)	Medium	High
Sunlight rejection	OBF filter (hardware)	N/A	Limited	Narrow beam only
Noise filter	HPF (electronic)	N/A	None	None
Outdoor use	Yes (with OBF + HPF)	Yes	No	Limited
Interception risk	Very low	High	Low	Low
EMI immunity	Complete	None	Complete	Complete
Cost	Low (LED + photodiode)	Moderate	Low	High

Table 2: Comparative analysis of SecureBeam LiFi against RF/WiFi, IrDA, and traditional FSO across key performance parameters.

The table makes a few things clear. SecureBeam has the highest security among LEDbased options because the IR signal physically cannot go anywhere outside the lineofsight path it stops at the receiver aperture. WiFi penetrates walls and can be intercepted from a car park; SecureBeam can't. The OBF + HPF combination is also what gives it a real edge over IrDA and FSO outdoors, since neither of those systems has a practical answer to solar noise. And because the whole thing runs on commodity IR LEDs and photodiodes, the cost stays well below FSO laser systems despite comparable data rates at ranges up to 100 metres.

7. Security Analysis

Security is the whole point of SecureBeam. This section looks at the four main threat dimensions eavesdropping, jamming, electromagnetic compatibility, and physical security and explains how the architecture handles each one.

7.1 Eavesdropping

To intercept the IR signal, an attacker needs to physically place a photodetector inside the beam path between the OBF and the receiver. The beam is narrow it diverges at about ±20° from the LED and the OBF aperture is only 2–5 cm across, so the beam footprint at the receiver is just a few centimetres wider than the detector itself. Anything that doesn't sit directly in that path picks up negligible signal. Better still, any physical intrusion into the beam breaks the signal at the receiver immediately, so the system detects its own eavesdropping attempts as a side effect of how optical communication works.

7.2 Jamming

Jamming with a bright light source is largely neutralised by the OBF. Only light in the 830–870 nm band at the correct angle of incidence can get through to the photodiode a broadband jammer gets rejected at OD4. A narrowband jammer at exactly 850 nm would need to be physically positioned right in front of the middle pole at the right incident angle, which means physical access to a location that can simply be fenced off or watched.

7.3 Electromagnetic Compatibility

SecureBeam generates zero RF emissions. That makes it immune to RF interference from nearby radar, highvoltage lines, or anything else and it means there's nothing for an RFbased sidechannel attack to pick up. For facilities that ban radio emissions entirely, it's one of the very few communication options that actually complies by design rather than by shielding.

7.4 Physical Security Zones

The threepole layout creates two inspectable security zones: the segment from the transmitter to the OBF (unfiltered beam zone) and the segment from the OBF to the receiver (clean beam zone). Both can be physically fenced, cameramonitored, or staffed. That layered physicalplusoptical security posture is something no RFbased system can replicate you simply can’t fence a radio signal.

8. Potential Applications

SecureBeam isn't a generalpurpose wireless system it's built for a specific set of deployment scenarios where security and RF restrictions take priority. Here are the most relevant ones.

8.1 Military and Defense Perimeters

Military checkpoints, border crossings, and forward operating bases often need to pass data securely between fixed positions tens to hundreds of metres apart. RF links in those environments are soft targets for jamming and interception; fiber optic runs are impractical when you're deploying fast. SecureBeam can be set up with three poles and a power supply, giving you a jamresistant, eavesdropresistant link suited to command and control data, biometric queries, and surveillance video.

8.2 HighSecurity Banking and Financial Infrastructure

Financial data centres and trading floors are increasingly required to go RFsilent to prevent data leakage through radio side channels. SecureBeam works well in that environment it can bridge server racks, trading terminals, or separate buildings across open courtyards without any cables and without breaking the RFsilent requirement.

8.3 Campus and Industrial InterBuilding Connectivity

Universities, industrial campuses, and corporate offices often need to connect buildings across courtyards or roads where trenching for cables is expensive and disruptive. SecureBeam is cheaper than FSO laser links and licensed microwave links, and IR LEDs don't require any spectrum licence you just set it up and it works.

8.4 Disaster Recovery Communication

When ground infrastructure is gone and the RF spectrum is saturated with emergency traffic, you need something that works independently of both. SecureBeam's hardware footprint is minimal three poles, an LED, a photodiode, a filter, and a power source which means it fits in an emergency kit and can be up and running quickly for command coordination links where it matters most.

9. Limitations and Future Scope

9.1 Current Limitations

SecureBeam isn’t perfect, and it’s worth being upfront about that. First, it needs strict lineofsight between all three poles. A passing vehicle, a person walking through, even a bird anything crossing the beam breaks the link. For applications where that’s unacceptable, you’d need a redundant parallel link or an automatic retry system. Second, heavy rain, dense fog, and dust storms all attenuate IR at 850 nm significantly, which can shrink the reliable range below what the design targets. Atmospheric turbulence above 100 metres adds signal fading that hurts BER. Third, OOK modulation caps the data rate at roughly 10 Mbps. If you need gigabit speeds, you’d have to move to OFDM or wavelengthdivision multiplexing, which means substantially more complexity and cost.

9.2 Future Research Directions

There are several directions worth pursuing from here. Swapping OOK for DMT or OFDM modulation could push data rates up by 10–100× without touching the threepole architecture itself all the extra complexity stays in the DSP stages. A feedback channel, possibly a lowbandwidth reverse IR link, could give the system adaptive link management: adjusting transmitter power and modulation order in real time based on atmospheric conditions, which would help a lot in fog and rain. Multibeam setups using LED arrays with multiple OBF apertures could create spatially multiplexed channels for throughput beyond what a single link can deliver the optical equivalent of MIMO. Further down the road, integrating quantum key distribution (QKD) over the same optical channel would push security to the unconditional level. And miniaturising the whole system onto drone platforms would enable mobile tactical LiFi links the drone deployment concept we considered and refined during this project’s development.

10. Conclusion

SecureBeam set out to solve a specific problem reliable LiFi communication in open outdoor conditions where sunlight destroys the signal and the threepole architecture is the answer we arrived at. By mounting the Optical Bandpass Filter on a dedicated elevated middle pole, the system physically separates solar noise rejection from the receiver electronics. The HighPass Filter on the receiver side then cleans up whatever DC bias and scintillation noise slips through. Together, those two stages produce a receiver SNR of roughly 30 dB under direct sunlight, giving a theoretical BER below 10^{-8} at 1 Mbps comfortably beating the 10^{-6} requirement.

On the security side, the system holds up well against the main threat vectors. The IR beam physically can't leave the lineofsight path, the system generates no RF emissions, and any physical intrusion into the beam breaks the link and alerts the receiver. All of that comes from commodity components IR LEDs, silicon PIN photodiodes, thinfilm bandpass filters, standard analog circuits which keeps the cost well below competing FSO laser systems.

The threepole layout itself transmitter, OBF filter, receiver is also a broader architectural idea worth noting: optical filter placement doesn't have to be fixed to the receiver. Treating it as a free variable in system design opens up options for other OWC architectures where the geometry of the noise source and the sensitivity of the detector create an opportunity for spatially separated filtering.

There's real demand for secure, RFimmune shortrange links in defence, finance, and critical infrastructure, and SecureBeam is positioned to serve it. The next steps are clear: higher data rates through advanced modulation, better atmospheric robustness through adaptive power control, and multibeam spatial multiplexing for scaling aggregate throughput.

References

[1] H. Haas, L. Yin, Y. Wang, and C. Chen, "What is LiFi?" Journal of Lightwave Technology, vol. 34, no. 6, pp. 1533–1544, 2016.

[2] D. Tsonev, S. Videv, and H. Haas, "Towards a 100 Gb/s visible light wireless access network," Optics Express, vol. 23, no. 2, pp. 1627–1637, 2015.

[3] Z. Ghassemlooy, W. Popoola, and S. Rajbhandari, Optical Wireless Communications: System and Channel Modelling with MATLAB. CRC Press, 2012.

[4] D. Karunatilaka, F. Zafar, V. Kalavally, and R. Parthiban, "LED Based Indoor Visible Light Communications: State of the Art," IEEE Communications Surveys & Tutorials, vol. 17, no. 3, pp. 1649–1678, 2015.

[5] S. D. Dissanayake and J. Armstrong, "Comparison of ACOOFDM, DCOOFDM and ADOOFDM in IM/DD Systems," Journal of Lightwave Technology, vol. 31, no. 7, pp. 1063–1072, 2013.

[6] A. Burton, H. Le Minh, Z. Ghassemlooy, E. Bentley, and C. Botella, "Experimental Demonstration of 50Mb/s Visible Light Communications Using 4 × 4 MIMO," IEEE Photonics Technology Letters, vol. 26, no. 9, pp. 945 –948, 2014.

[7] IEEE Standard 802.11bb, "IEEE Standard for Local and Metropolitan Area Networks Light Communications," IEEE, 2023.

[8] P. Brandl, R. Swoboda, and H. Zimmermann, "Fully integrated optical receiver for 1 Gb/s visible light communication," IEEE Photonics Technology Letters, vol. 22, no. 22, pp. 1665–1667, 2010.